

Supplementary Information for

Acceleration of an Antarctic outlet glacier driven by surface meltwater input to the base

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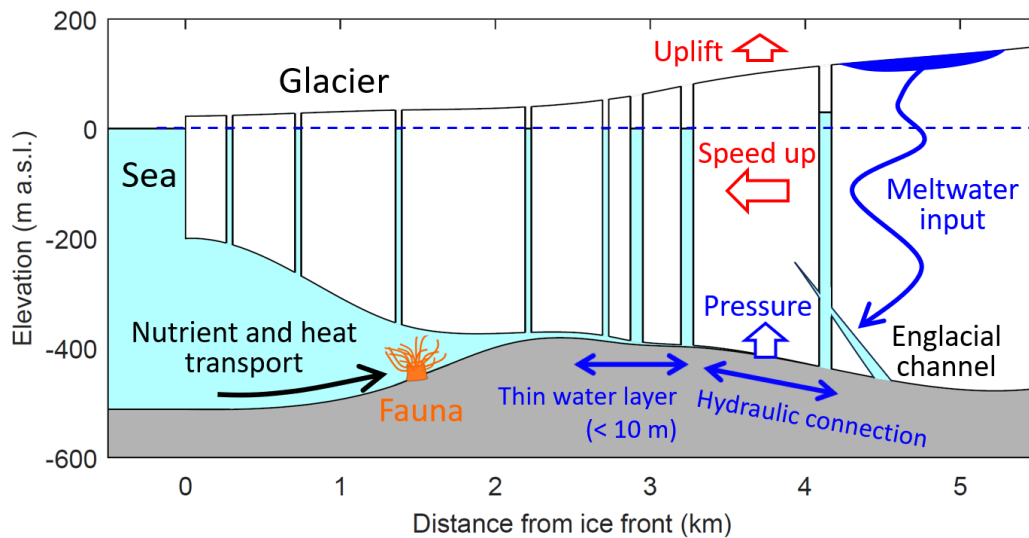
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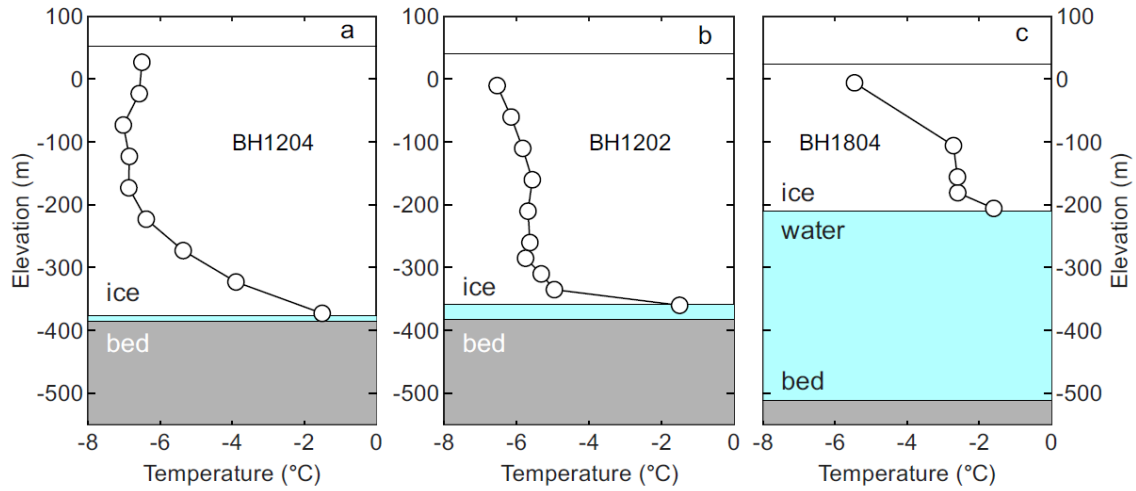
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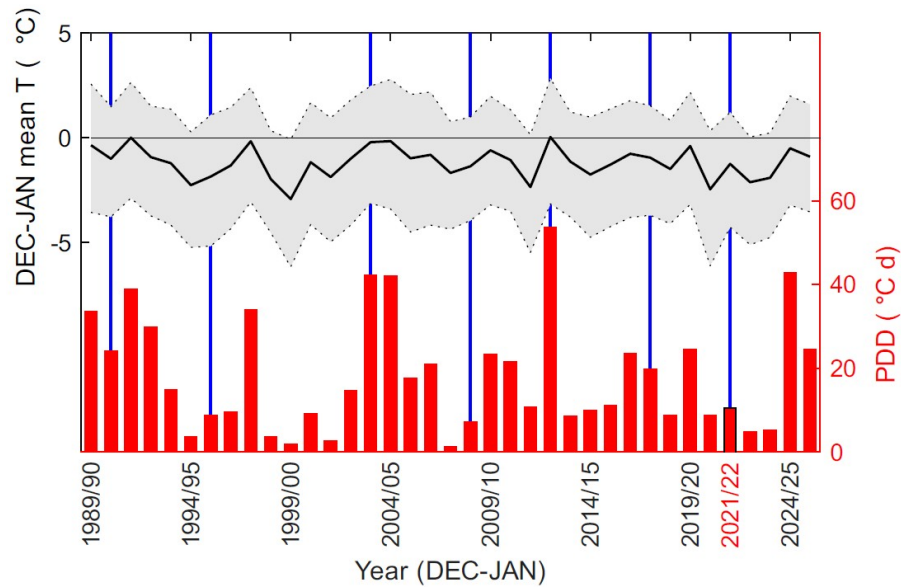
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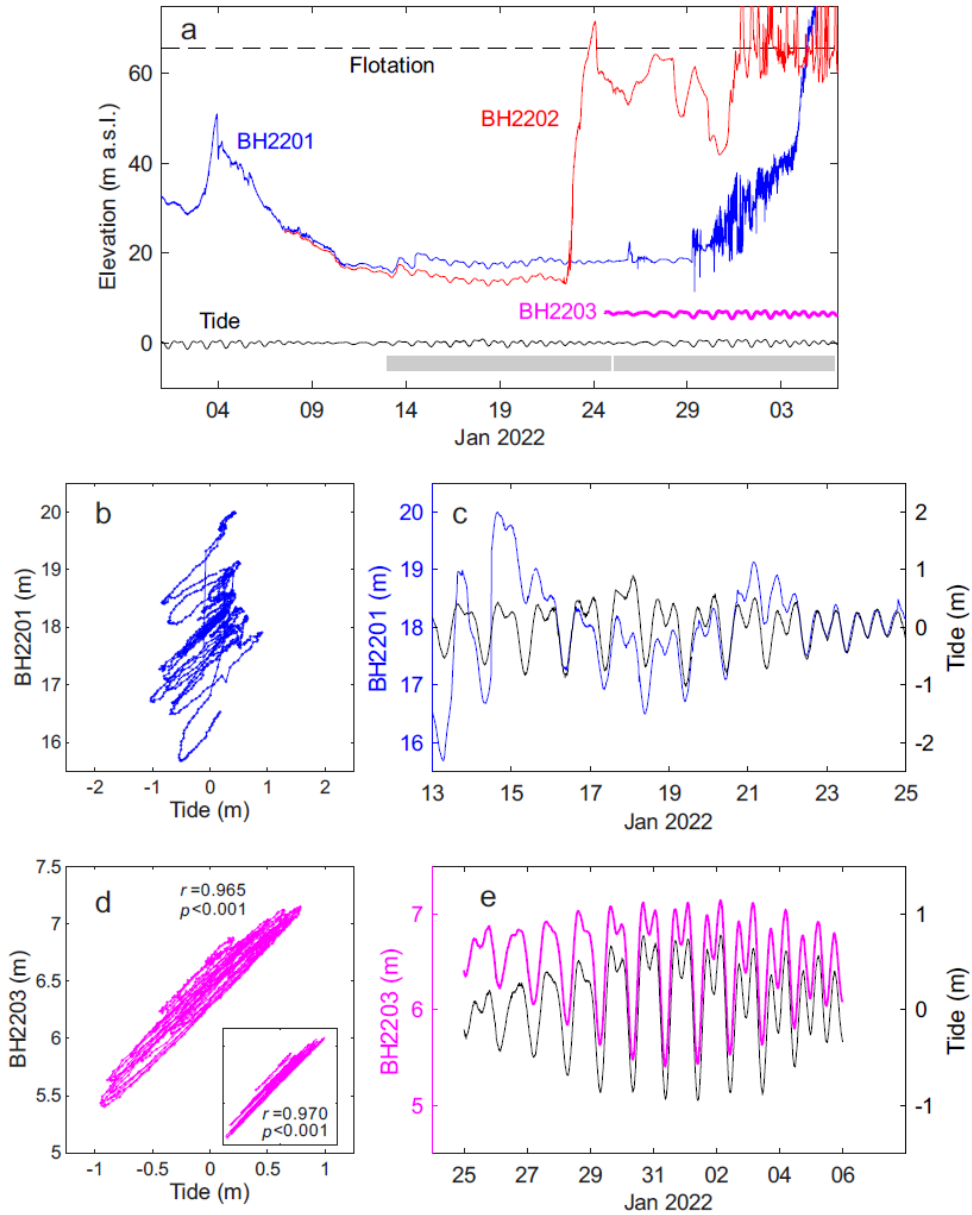
Supplementary Figure 1. Schematic diagram showing findings of this study. Hot-water drilling and borehole measurements revealed several key findings: glacier acceleration driven by meltwater input to the base; a hydraulic connection to the sub-shelf cavity; and the presence of fauna in a thin water layer observed near the grounding line.



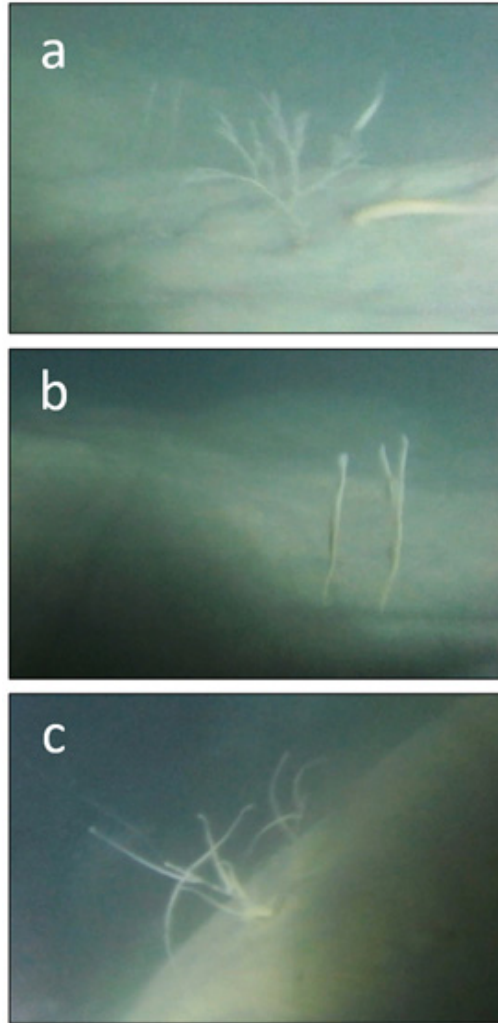
Supplementary Figure 2. Englacial temperature distributions. Temperature-depth profiles measured in boreholes (a) BH1204, (b) BH1202 and (c) BH1804. The markers show the locations of thermistor sensors.



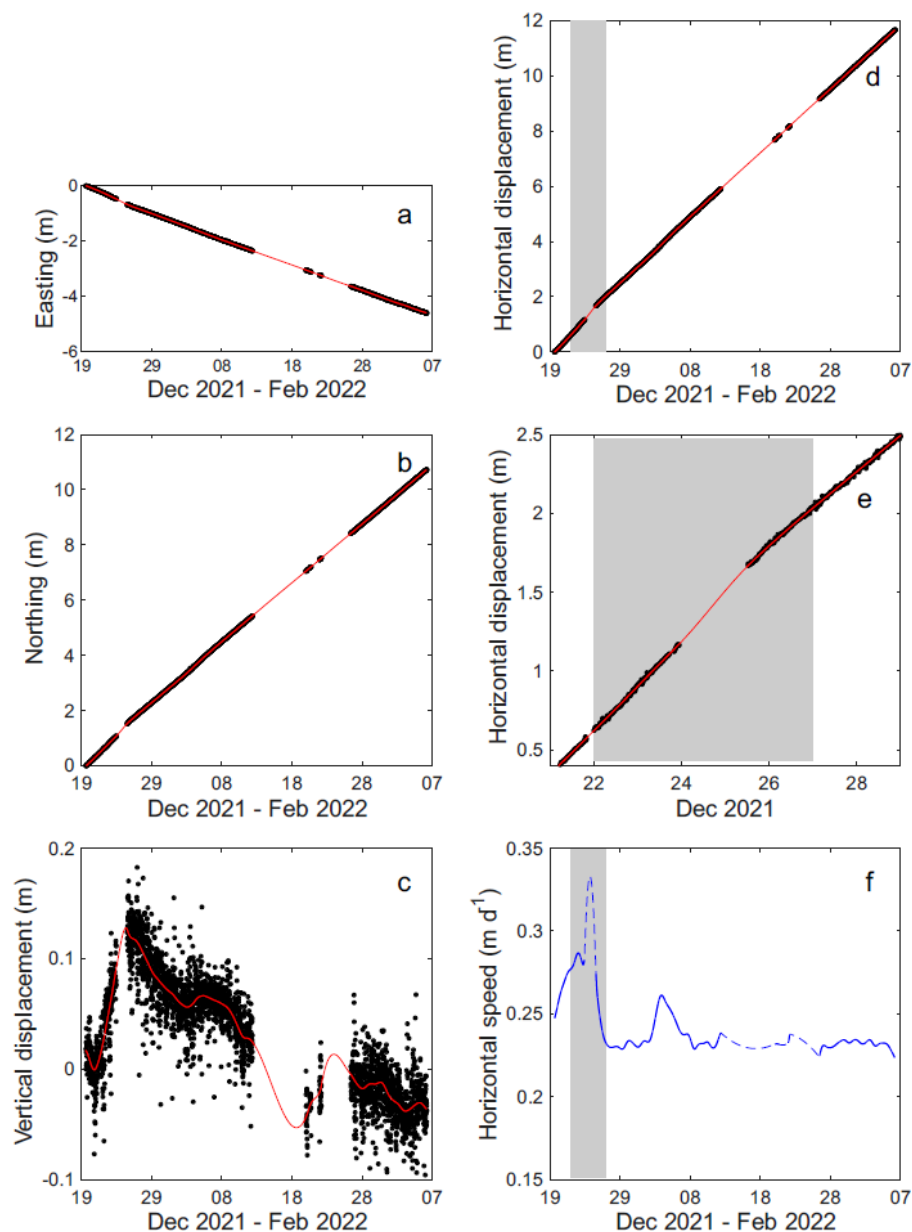
Supplementary Figure 3. Summer (December and January) meteorological conditions at Syowa Station from 1989/20 to 2025/26. Mean air temperature (black) with mean daily minimum/maximum temperatures (dotted), the sum of positive degree days (red), and rain events (blue).



Supplementary Figure 4. Borehole water levels and ocean tides. (a) Water level in boreholes BH2201 (blue), BH2202 (red) and BH2203 (magenta). The solid black line indicates the ocean tides measured at Syowa Station. The dashed line is the flotation water level at BH2201/02. The grey bands indicate the periods covered by (b–c) and (d–e). (b) Scatter and (c) time series plots of tide and water level in borehole BH2201, and (d–e) plots of tide and water level in borehole BH2203. The inset in (d) shows that the correlation improves by introducing a time lag of 0.25 hr, i.e. BH2203 level (t) v.s. tide ($t+0.25$ hr). Correlation coefficients (r) and p -values are indicated.



Supplementary Figure 5. Sub-shelf images taken at (D–F) in Fig. 1d. (a–c) Borehole camera images, showing fauna observed on the seafloor below BH2203. Images are ~200 mm wide.



Supplementary Figure 6. Filtering the positioning data and ice speed estimation for the data gaps. (a–c) GNSS positioning data obtained at GNSS1 for easting, northing and vertical coordinates (black) and the results of the filtering and spline interpretation (red). (d) Horizontal displacement computed from the easting and northing coordinates before (black) and after the filtering and interpretation (red). (e) Enlarged plot of (d) for the first speed-up (Period I) highlighted with the grey band. (f) Ice speed obtained by the differentiation of the horizontal displacement. The dashed lines correspond to the data gap periods.